

Block (1/3)

The Finite Element Method for the Diffusion Equation in 1D

EE4001-IC - FEM For EE Applications

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One-Dimensional Problem to Solve (1/3)

Same as Before: Boundary Value Problem
Second Order Differential Equations

- **given** $x \in \Omega = (0, 1)$ with left and right boundary point
- **given**: $f(x)$ given function and α given number
- **find**: $u(x)$ such that

$$-u''(x) = f(x) \text{ for } 0 < x < 1 \text{ (differential equation on } \Omega)$$

$$u(x = 0) = 0 \text{ (Dirichlet boundary condition in } x = 0)$$

$$\frac{du}{dx}(x = 1) = \alpha \text{ (Neumann boundary condition in } x = 1)$$

- differential equation **not** valid in $x = 0$ or $x = 1$
boundary conditions **are** valid in $x = 0$ or $x = 1$

One-Dimensional Problem to Solve (2/3)

Residual function $r(x)$

- **definition:** $r(x) = u''(x) + f(x)$
- **quality of the solution:**
small (large) residual is indication of good (poor) approximation
- **solve** $-u''(x) = f(x) + \text{b.c.}$ equivalent to
- **find** $u(x)$ such that $r(x) = 0$ and $u(x)$ satisfies b.c.

One-Dimensional Problem to Solve (3/3)

Reference Solutions

- **analytical**: integrate twice $f(x)$ and use boundary conditions to determine constants of integration. Or guess repeated guessing. Or choose a reference solution and determine $f(x)$ and boundary conditions accordingly.
- **symbolically**: use e.g. the function `dsolve()` in the SymPy.jl package;
- **numerical**: use a existing package e.g. the function `solve_bvp()` in the DifferentialEquations.jl package, Ferrite.jl, Gridap.jl, ApproxFun.jl or otherwise

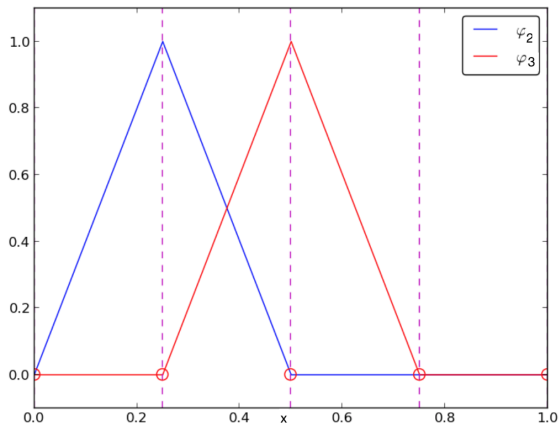
Mesh and Shape Functions (1/20)

$$\text{Mesh } \Omega^h = \{x_i \mid 1 \leq i \leq N + 1\}$$

- create mesh on $\Omega = (0, 1)$
- $(N + 1)$ points or **nodes**: x_i for $1 \leq i \leq N + 1$
- N intervals or **elements**: $e_i = [x_i, x_{i+1}]$ for $1 \leq i \leq N$
- element size $h_i = x_{i+1} - x_i$
- mesh uniform in case that h_i is independent of i
non-uniform otherwise
- **Let op (Dutch) - Dhyaan (Hindi) - Zhuyi li (Chinese) - Alaintibah (Arabic):**
N elements e_i - (N+1) nodes x_i at which 2 on the boundary

Mesh and Shape Functions (2/20)

Graphs of shape functions



Mesh and Shape Functions (3/20)

Definition of linear lagrange shape function $\phi_i(x)$

- linear Lagrange interpolation function
- for $x_{i-1} \leq x \leq x_i$ (left of x_i): $\phi_i(x) = \frac{x - x_{i-1}}{x_i - x_{i-1}}$ (positive slope)
- for $x_i \leq x \leq x_{i+1}$ (right of x_i): $\phi_i(x) = \frac{x_{i+1} - x}{x_{i+1} - x_i}$ (negative slope)
- for $x \leq x_{i-1}$ or $x \geq x_{i+1}$: $\phi_i(x) = 0$
- each node x_i (including 2 boundary nodes) has its $\phi_i(x)$
- mesh defines in total $N+1$ functions $\phi_i(x)$
- $\phi_i(x_j) = \delta_{ij}$ (see figure)

Mesh and Shape Functions (4/20)

Mesh Terminology

1D Domains Ω

- two nodes x_i
- interval $e_i = [x_i, x_{i+1}]$
- interval length $h_i = x_{i+1} - x_i$

2D Domains Ω

- three nodes x_i
- triangle formed by x_i
- area of triangle

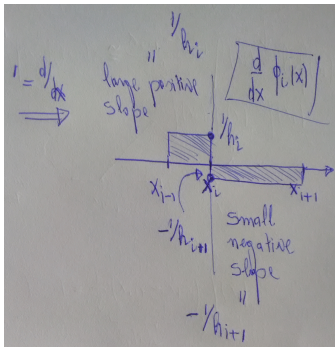
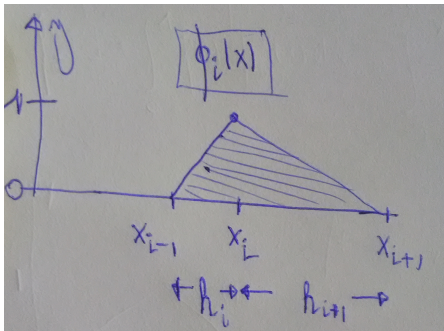
Mesh and Shape Functions (5/20)

Shape functions

- the mesh **determines** the set of shape functions
- construction carries over to **non-uniform** meshes
- **one** element $e_i = [x_i, x_{i+1}]$ "sees" **two** basis functions
 - left node x_i has basis function $\phi_i(x)$
 - right node x_{i+1} has basis function $\phi_{i+1}(x)$
 - all other basis function $\phi_j(x)$ with $j \neq i$ or $j \neq i + 1$ are zero on e_i

Mesh and Shape Functions (6/20)

Shape function $\phi_i(x)$ (left) - derivative $\phi'_i(x) = \frac{d}{dx}\phi_i(x)$ (right)



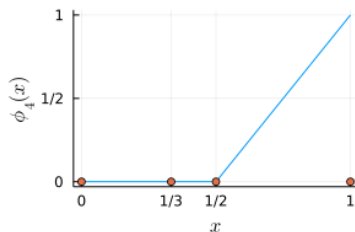
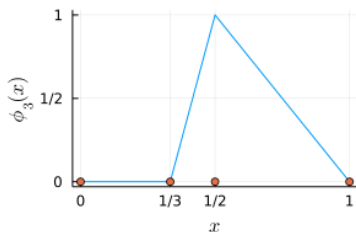
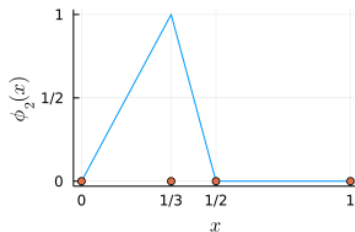
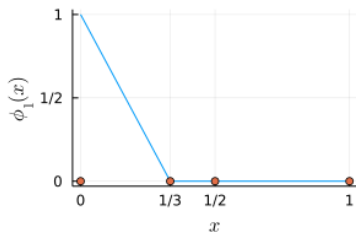
Mesh and Shape Functions (7/20)

Exercises on linear Lagrangian shape functions

- assume a mesh with the four nodes
 $x_1 = 0$, $x_2 = 1/3$, $x_3 = 1/2$ and $x_4 = 1$;
- plot the mesh and label the nodes x_1 , x_2 , x_3 and x_4 ;
- plot the shape functions $\phi_1(x)$, $\phi_2(x)$, $\phi_3(x)$ and $\phi_4(x)$;
- plot the first derivative of the shape functions $\frac{d\phi_1(x)}{dx}$, $\frac{d\phi_2(x)}{dx}$, $\frac{d\phi_3(x)}{dx}$ and $\frac{d\phi_4(x)}{dx}$;
- plot the second derivative of the shape functions $\frac{d^2\phi_1(x)}{dx^2}$, $\frac{d^2\phi_2(x)}{dx^2}$, $\frac{d^2\phi_3(x)}{dx^2}$ and $\frac{d^2\phi_4(x)}{dx^2}$;

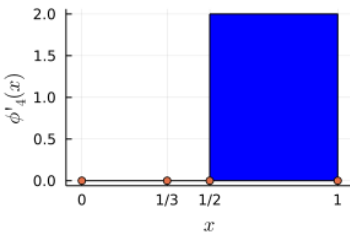
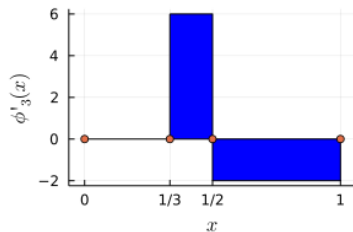
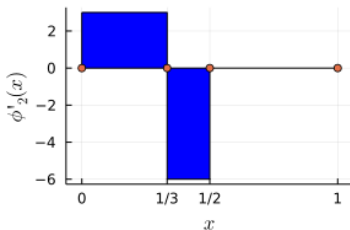
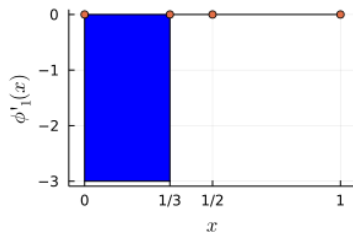
Mesh and Shape Functions (8/20)

Solution (1/2): Shape Functions



Mesh and Shape Functions (9/20)

Solution (2/2): Shape Function Derivative



Mesh and Shape Functions (10/20)

Linear Combination of Shape Functions

- set of functions $\{\phi_i(x) | 1 \leq i \leq N+1\}$
- **linear combinations** of these functions $\phi_i(x)$ can be made
- $V_0^h(\Omega)$: function space defined by all linear combinations

$$V_0^h(\Omega) = \text{span}\{\phi_1(x), \dots, \phi_{N+1}(x)\}$$

- $u^h(x) \in V_0^h(\Omega)$: there exists coordinates $c_1, \dots, c_{N+1}$ such that

$$u^h(x) = c_1 \phi_1(x) + \dots + c_{N+1} \phi_{N+1}(x)$$

Mesh and Shape Functions (11/20)

Analogy with Geometry

- geometry in 2D plane: $\mathbf{x} = x \mathbf{i} + y \mathbf{j}$
- x and y coordinates - $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$ basis vectors
- geometry in 3D space: $\mathbf{x} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$
- x , y and z : coordinates
- $\mathbf{i} = (1, 0, 0)$, $\mathbf{j} = (0, 1, 0)$ and $\mathbf{k} = (0, 0, 1)$: basis vectors

Mesh and Shape Functions (12/20)

Analogy with Fourier Analysis

- $f(x)$: given function on domain $x \in \Omega = (0, 1)$
- $f^N(x)$: approximation of $f(x)$ with all harmonics up to order N

$$f^N(x) = a_0 + \sum_{k=1}^N a_k \cos(k \pi x) + b_k \sin(k \pi x)$$

- a_0 , a_k and b_k for $1 \leq k \leq N$: **coordinates** or **mode amplitudes**
- 1 , $\cos(k \pi x)$ and $\sin(k \pi x)$: Fourier modes or **basis functions**
- Fourier modes form an **orthonormal** set of functions

Mesh and Shape Functions (13/20)

Expressions for the Fourier Mode Amplitudes a_0 , a_k and b_k

- $f^N(x) = a_0 + \sum_{k=1}^N a_k \cos(k \pi x) + b_k \sin(k \pi x)$
- $a_0 = \langle f(x), 1 \rangle$
- $a_k = \frac{1}{\sqrt{2}} \langle f(x), \cos(k \pi x) \rangle = \frac{1}{\sqrt{2}} \int_0^1 f(x) \cos(k \pi x) dx$
- $b_k = \frac{1}{\sqrt{2}} \langle f(x), \sin(k \pi x) \rangle = \frac{1}{\sqrt{2}} \int_0^1 f(x) \sin(k \pi x) dx$
- see textbook on Fourier series

Mesh and Shape Functions (14/20)

Analogy with Separation of Variables

- $f(x)$: given function on domain $x \in \Omega = (0, 1)$ as before
- assume λ_k (eigenvalue) and $u^k(x)$ (eigenfunction) such that

$$-\left[u^k(x)\right]'' = \lambda_k u^k(x) \text{ plus boundary conditions}$$

- then as before

$$f(x) = \sum_{k=1}^{\infty} a_k u^k(x)$$

- see textbook on separation of variables

Mesh and Shape Functions (15/20)

Application to Finite Elements

- $u(x)$: **exact** solution of the boundary value problem
- $u^h(x)$: finite element **approximation** to $u(x)$ computed on Ω^h
- $u^h(x) \in V_0^h(\Omega) = \text{span}\{\phi_1(x), \dots, \phi_{N+1}(x)\}$
- expansion of $u^h(x)$ as linear combination of shape function

$$u^h(x) = c_1 \phi_1(x) + \dots + c_{N+1} \phi_{N+1}(x)$$

- $c_1, \dots, c_{N+1}$ **coordinates** - $\phi_1(x), \dots, \phi_{N+1}(x)$ **basis functions**
- non-uniform meshes can be accommodated

Mesh and Shape Functions (16/20)

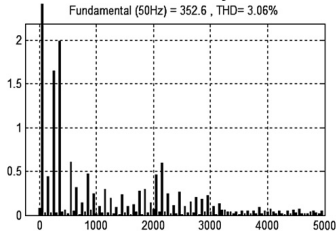
Expressions for the coordinate c_i

- $u^h(x) = c_1 \phi_1(x) + \dots + c_{N+1} \phi_{N+1}(x)$
- c_i computed by $\int_0^1 [-u^h(x)]'' \phi_i(x) dx = \int_0^1 f(x) \phi_i(x) dx$
- more details later
- compare with the expression for the Fourier modes a_0 , a_k and b_k

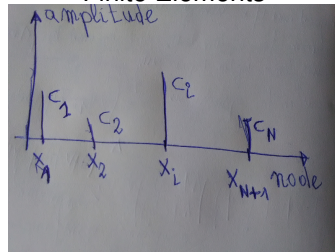
Mesh and Shape Functions (17/20)

Comparing Fourier Analysis and Finite Elements

Fourier Analysis



Finite Elements



Mesh and Shape Functions (18/20)

Exercises

- given non-uniform mesh on Ω with 4 nodes $x = 0$, $x = 1/3$, $x = 1/2$ and $x = 1$
- plot first order Lagrange shape functions $\phi_1(x)$, $\phi_2(x)$, $\phi_3(x)$ and $\phi_4(x)$
- plot linear combinations of $\phi_1(x)$, $\phi_2(x)$, $\phi_3(x)$ and $\phi_4(x)$
- assume a non-uniform mesh on Ω with $(N+1)$ nodes
- plot $\phi_i(x) + \phi_{i+1}(x)$ on e_i

Mesh and Shape Functions (19/20)

Quadratic Lagrange Shape Functions

- given non-uniform mesh on Ω with 4 nodes $x = 0$, $x = 1/3$, $x = 1/2$ and $x = 1$
- plot second order Lagrange shape functions $\phi_1(x)$, $\phi_2(x)$, $\phi_3(x)$ and $\phi_4(x)$, their derivatives and linear combinations of these shape functions
- more ...

Mesh and Shape Functions (20/20)

h vs. p refinement

- here soon

Weak or Variational Form (1/22)

When is $g(x) = 0$ on the domain Ω in Strong Form?

- assume $g(x)$ scalar function in x with domain $\Omega = (0, 1)$ (as $u(x)$ and $f(x)$ before)
- **analytically** - without any numerical approximation:

$g(x) = 0$ if and only if for all $x \in \Omega$ we have that $g(x) = 0$

$$g(x) = 0 \text{ in continuous strong form} \Leftrightarrow \forall x \in \Omega : g(x) = 0$$

- as expected
- two functions $g_1(x)$ and $g_2(x)$ are equal in strong form on Ω iff the difference $g_1(x) - g_2(x)$ is equal to zero in strong form on Ω

Weak or Variational Form (2/22)

When is $g(x) = 0$ on the mesh Ω^h in Strong Form?

- **numerically:** $g(x_i) = 0 \quad \forall x_i \in \Omega^h = \{x_j | 1 \leq j \leq N+1\}$

lots of space to choose $g(x)$ for $x \neq x_i$

e.g. $g(\frac{x_i+x_{i+1}}{2}) = \text{whatever-you-please}$

- as the mesh Ω^h is refined, the discrepancy between $g(x) = 0$ on the domain Ω and $g(x) = 0$ on the mesh Ω^h decreases

Weak or Variational Form (3/22)

Discussion of Equal to Zero in Strong Form

- **analytically:** $g(x) = 0 \quad \forall x \in \Omega = (0, 1)$

no space to choose $g(x)$ for input x

- **numerically:** $g(x_i) = 0 \quad \forall x_i \in \Omega^h = \{x_j | 1 \leq j \leq N + 1\}$

lots of space to choose $g(x)$ for $x \neq x_i$

e.g. $g(\frac{x_i + x_{i+1}}{2}) = \text{whatever-you-please}$

Weak or Variational Form (4/22)

Example of Equal to Zero in Strong Form

- **choose:** $g(x) = r(x) = u''(x) + f(x)$ (residual function)

- **analytically:**

$$g(x) = 0 + bc \Leftrightarrow -u''(x) = f(x) + bc \Leftrightarrow \text{problem to solve}$$

- **numerically:**

$$g(x_i) = 0 + bc \Leftrightarrow -u''(x_i) = f(x_i) + bc \Leftrightarrow \text{finite diff. method}$$

Weak or Variational Form (5/22)

When is $g(x) = 0$ on the domain Ω in Weak Form?

- **first idea** using averaging over Ω

$$g(x) = 0 \text{ in weak form} \Leftrightarrow \int_{\Omega} g(x) dx = \int_0^1 g(x) dx = 0$$

- strong equal zero **implies** weak equal zero

$$g(x) = 0 \Rightarrow \int_0^1 g(x) dx = 0$$

- converse is **false** (try e.g. $g(x) = x - 1/2$)
- hence the terminology
- can equal to zero in strong form be recovered from equal to zero in weak form?

Weak or Variational Form (6/22)

When is $g(x) = 0$ on the domain Ω in Weak Form (cont'd)?

- **refined idea** using weighted averaging over Ω

$$g(x) = 0 \text{ in weak form} \Leftrightarrow \int_{\Omega} g(x) v(x) dx = \int_0^1 g(x) v(x) dx = 0$$

- $v(x)$ is a weighting function
- in previous slide $v(x) = 1$ was chosen
- What form should $v(x)$ have? How many $v(x)$ are there?

Weak or Variational Form (7/22)

When is $g(x) = 0$ on the mesh Ω^h in Weak Form?

Inspiration from Linear Algebra

- n -dimensional vector $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$
- unit-vector $\mathbf{i}$ i -th component equal to one, all other components equal to zero
- $\mathbf{v} = \mathbf{0} \Leftrightarrow v_i = 0 \quad 1 \leq i \leq n \quad n+1$ equations indexed by i
- $\mathbf{v} = \mathbf{0} \Leftrightarrow \mathbf{v}^T \mathbf{i} = \langle \mathbf{v}, \mathbf{i} \rangle = 0 \quad 1 \leq i \leq n$
- how to bring this to finite elements?

Weak or Variational Form (8/22)

Applied to Finite Elements

- $N + 1$ basis functions $\phi_i(x)$ defined by the mesh Ω^h
- inner product: $\langle g(x), \phi_i(x) \rangle = \int_0^1 g(x) \phi_i(x) dx$
- $\langle g(x), \phi_i(x) \rangle$ coordinate of $g(x)$ along the basis function $\phi_i(x)$
- **numerically**: essential in remainder of the course

$$g(x) = 0 \text{ in discrete weak form} \Leftrightarrow \forall x_i \in \Omega^h : \langle g(x), \phi_i(x) \rangle = 0$$

again $N + 1$ equations indexed by i

- **analytically**: see later

Weak or Variational Form (9/22)

Applied to Finite Elements

- choose $g(x) = r(x) = u''(x) + f(x)$
- enforce $g(x) = 0$ plus boundary conditions in discrete weak form

$$\langle g(x), \phi_i(x) \rangle = 0 \quad \text{for } 1 \leq i \leq N+1$$

$$\Leftrightarrow \int_0^1 g(x) \phi_i(x) dx = 0 \quad \text{for } 1 \leq i \leq N+1$$

$$\Leftrightarrow \int_0^1 -u''(x) \phi_i(x) dx = \int_0^1 f(x) \phi_i(x) dx \quad \text{for } 1 \leq i \leq N+1$$

Weak or Variational Form (10/22)

Applied to Finite Elements

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Weak or Variational Form (11/22)

Terminology

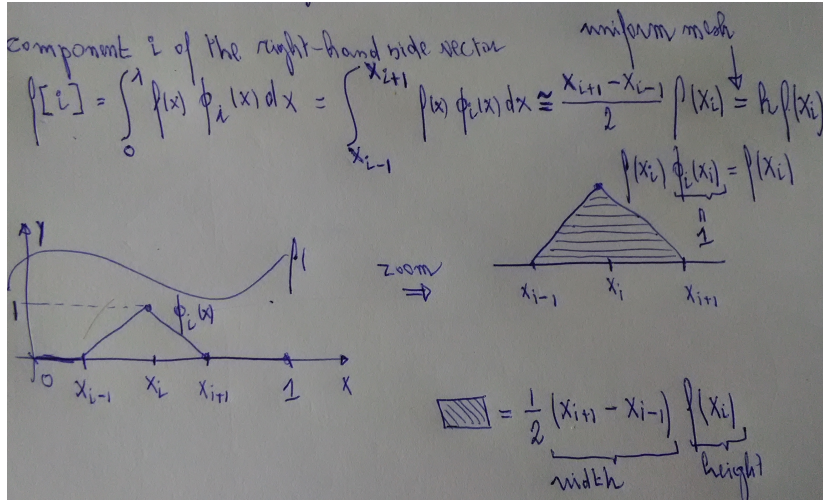
- $f(x)$: source term or right-hand side in the differential equation
- α : value imposed on the derivative
- $\phi_i(x)$: basis function centered on $x_i \in \Omega^h$
- f_i : i -th component of the right-hand side vector $\mathbf{f}$
- relationship (see previous slide)

$$f_i = \int_0^1 f(x) \phi_i(x) dx$$

Weak or Variational Form (12/22)

$$\begin{aligned}f_i &= \int_0^1 f(x) \phi_i(x) dx = \int_{x_{i-1}}^{x_{i+1}} f(x) \phi_i(x) dx \quad (\text{restrict to } e_{i-1} \cup e_i) \\&= \int_{x_{i-1}}^{x_i} f(x) \phi_i(x) dx + \int_{x_i}^{x_{i+1}} f(x) \phi_i(x) dx \quad (\text{separate } e_{i-1} \text{ and } e_i) \\&\approx \frac{h_{i-1}}{2} [f(x_{i-1}) \phi_i(x_{i-1}) + f(x_i) \phi_i(x_i)] + \frac{h_i}{2} [f(x_i) \phi_i(x_i) + f(x_{i+1}) \phi_i(x_{i+1})] \\&\quad (\text{use trapezoidal rule twice}) \\&= \frac{h_{i-1}}{2} [f(x_{i-1}) 0 + f(x_i) 1] + \frac{h_i}{2} [f(x_i) 1 + f(x_{i+1}) 0] \\&\quad (\text{use properties of } \phi_i(x)) \\&= \frac{h_{i-1} + h_i}{2} f(x_i) \\&= h f(x_i) \quad (\text{in case of uniform mesh})\end{aligned}$$

Weak or Variational Form (13/22)



Weak or Variational Form (14/22)

Treatment of the Dirichlet and Neumann boundary conditions

- **Dirichlet** boundary condition

fix $u(x)$ - grounding:

same as for Finite Difference method

- **Neumann** boundary condition

fix $\frac{d}{dx} u(x)$ - imposed symmetry or flux:

different from Finite Difference method - details later

Weak or Variational Form (15/22)

Wish - Desire - Aim - Ambition?

- manipulate $-\int_0^1 u''(x) \phi_i(x) dx$
- challenge: $u(x)$ is unknown
- observe: $u''(x)$ has double prime - $\phi_i(x)$ has no prime

Weak or Variational Form (16/22)

Derivative: Integration by Parts in one variable (1/2)

- **fundamental theorem of calculus:**

assume $0 < x < 1$ or $x \in \Omega = (0, 1)$ and $F'(x) = \frac{dF(x)}{dx}$

$$\int_{\Omega} F'(x) dx = \int_0^1 F'(x) dx = [F(x)]_0^1 = F(1) - F(0)$$

- **choose** $F(x) = \frac{du}{dx} v(x)$ (motivation on next slide)

observe $u(x)$ has prime and $v(x)$ has no prime

- then $F'(x) = \frac{d}{dx} \left(\frac{du}{dx} v(x) \right) = u''(x) v(x) + u'(x) v'(x)$

Weak or Variational Form (17/22)

Derivative: Integration by Parts in one variable (2/2)

- from $\int_0^1 F'(x) dx = [F(x)]_0^1$ we obtain that

$$\int_0^1 [u''(x) v(x) + u'(x) v'(x)] dx = [u'(x) v(x)]_0^1$$

- after rearranging terms

$$\boxed{- \int_0^1 u''(x) v(x) dx = \int_0^1 u'(x) v'(x) dx - [u'(x) v(x)]_0^1}$$

LHS: $u(x)$ has double prime and $v(x)$ has no prime

RHS: both $u(x)$ and $v(x)$ have one prime - additional fudge term
(used later)

Weak or Variational Form (18/22)

Integration: Quadrature by Trapezoidal Rule

- trapezoidal rule: $0 \leq a < b \leq 1$

$$\int_a^b g(x) dx \approx \frac{b-a}{2} [g(a) + g(b)]$$

- Simpson rule: $0 \leq a < b \leq 1$

$$\int_a^b g(x) dx \approx \frac{b-a}{6} \left[g(a) + 4g\left(\frac{a+b}{2}\right) + g(b) \right]$$

- typically applied to $a = x_i$ and $b = x_{i+1}$

Weak or Variational Form (19/22)

Apply Integration by Parts on Weak Formulation

- earlier we set the residual $r(x) = u''(x) + f(x)$ to zero in weak form and arrived at

$$\int_0^1 -u''(x) \phi_i(x) dx = \int_0^1 f(x) \phi_i(x) dx \quad \text{for } 1 \leq i \leq N+1$$

- apply integration by part to the LHS

$$\int_0^1 u'(x) \phi_i'(x) dx = \int_0^1 f(x) \phi_i(x) dx + [u'(x) \phi_i(x)]_0^1 \quad \text{for } 1 \leq i \leq N+1$$

observe: minus sign in LHS disappeared

- first order derivative on both $u(x)$ and $\phi_i(x)$

Weak or Variational Form (20/22)

- boundary conditions: $u(x = 0) = 0$ and $u'(x = 1) = \alpha$
- boundary term in the RHS of the weak form

$$\begin{aligned}[u'(x) \phi_i(x)]_0^1 &= u'(x = 1) \phi_i(x = 1) - u'(x = 0) \phi_i(x = 0) \\ &= \alpha \phi_i(x = 1) - (\text{no information given}) \phi_i(x = 0)\end{aligned}$$

- Dirichlet and Neumann boundary conditions treated differently
- for now we assume that

$$[u'(x) \phi_i(x)]_0^1 = \alpha \phi_i(x = 1) - 0 = \alpha \phi_i(x = 1)$$

Weak or Variational Form (21/22)

Discrete Weak Form Becomes

$$\int_0^1 u'(x) \phi_i'(x) dx = \int_0^1 f(x) \phi_i(x) dx + \alpha \phi_i(x=1) \quad \text{for } 1 \leq i \leq N+1$$

- assume $u(x)$ approximate by $u^h(x)$ where $u^h(x) = \sum_{j=1}^{N+1} c_j \phi_j(x)$
- thus $u'(x)$ approximate by $[u^h(x)]' = \sum_{j=1}^{N+1} c_j \phi_j'(x)$
- then for $1 \leq i \leq N+1$

$$\sum_{j=1}^{N+1} \int_0^1 \phi_j'(x) \phi_i'(x) dx c_j = \int_0^1 f(x) \phi_i(x) dx + \alpha \phi_i(x=1)$$

Weak or Variational Form (22/22)

Discrete Weak Form Becomes

- for $1 \leq i \leq N + 1$

$$\sum_{j=1}^{N+1} \int_0^1 \phi_i'(x) \phi_j'(x) dx c_j = \int_0^1 f(x) \phi_i(x) dx + \alpha \phi_i(x=1)$$

- can be written in the form: for $1 \leq i \leq N + 1$

$$\sum_{j=1}^{N+1} A_{ij} c_j = f_i$$

- and thus as a $N + 1$ by $N + 1$ linear system

$$\mathbf{A} \mathbf{c} = \mathbf{f}$$

- index i counts equations - index j counts unknowns

Linear System Formulation (1/8)

Expression for Matrix and Vector Elements

- Matrix elements:

$$A_{ij} = \int_0^1 \phi'_i(x) \phi'_j(x) dx \text{ for } 1 \leq i, j \leq N + 1$$

- Vector elements:

$$f_i = \int_0^1 f(x) \phi_i(x) dx + \alpha \phi_i(x = 1) \text{ for } 1 \leq i \leq N + 1$$

Linear System Formulation (2/8)

Properties of Matrix A

- diagonal of A is **positive** - $A_{ii} > 0$

$$A_{ii} = \int_0^1 \underbrace{\phi'_i(x) \phi'_i(x)}_{>0} dx > 0$$

- A is **symmetric** - $A = A^T$

$$A_{ij} = \int_0^1 \phi'_i(x) \phi'_j(x) dx = \int_0^1 \phi'_j(x) \phi'_i(x) dx = A_{ji}$$

- **off-diagonal** of A is negative

$$A_{i,i+1} = \int_0^1 \phi'_i(x) \phi'_{i+1}(x) dx = \int_{x_i}^{x_{i+1}} \underbrace{\phi'_i(x)}_{<0} \underbrace{\phi'_{i+1}(x)}_{>0} dx < 0$$

- A is **tri-diagonal** and thus sparse
- A many other cool properties $\Rightarrow$ **fast solvers** for $A\mathbf{c} = \mathbf{f}$ exist

Linear System Formulation (3/8)

Remember that

- length of element $e_i = [x_i, x_{i+1}]$ is denoted by $h_i = x_{i+1} - x_i$
- length of element $e_{i-1} = [x_{i-1}, x_i]$ is denoted by $h_{i-1} = x_i - x_{i-1}$
- slope of $\phi_i(x)$ on e_i is equal to $\frac{-1}{h_i}$
- slope of $\phi_{i+1}(x)$ on e_i is equal to $\frac{1}{h_i}$

Linear System Formulation (4/8)

Diagonal Matrix Elements A_{ii}

$$\begin{aligned} A_{ii} &= \int_0^1 \phi'_i(x) \phi'_i(x) dx = \int_{x_{i-1}}^{x_{i+1}} \phi'_i(x) \phi'_i(x) dx \quad (\text{restrict to } e_{i-1} \cup e_i) \\ &= \int_{x_{i-1}}^{x_i} \phi'_i(x) \phi'_i(x) dx + \int_{x_i}^{x_{i+1}} \phi'_i(x) \phi'_i(x) dx \quad (\text{separate } e_{i-1} \text{ and } e_i) \\ &= \int_{x_{i-1}}^{x_i} \left(\frac{1}{h_{i-1}} \right)^2 dx + \int_{x_i}^{x_{i+1}} \left(\frac{-1}{h_i} \right)^2 dx \quad (\text{use information on derivatives}) \\ &= h_{i-1} \left(\frac{1}{h_{i-1}} \right)^2 + h_i \left(\frac{1}{h_i} \right)^2 \quad (\text{integration of constant integrands}) \\ &= \frac{1}{h_{i-1}} + \frac{1}{h_i} \quad (\text{in case of non-uniform mesh}) \\ &= \frac{2}{h} \quad (\text{in case of uniform mesh for which all } h_i \text{ are equal}) \end{aligned}$$

Linear System Formulation (5/8)

Upper-Diagonal Matrix Elements

$$\begin{aligned} A_{i,i+1} &= \int_0^1 \phi'_i(x) \phi'_{i+1}(x) dx = \int_{x_i}^{x_{i+1}} \phi'_i(x) \phi'_{i+1}(x) dx \quad (\text{restrict to } e_i) \\ &= \int_{x_i}^{x_{i+1}} \frac{-1}{h_i} \frac{1}{h_i} dx \quad (\text{use information on derivatives}) \\ &= h_i \frac{-1}{h_i} \frac{1}{h_i} \quad (\text{integration of constant integrands}) \\ &= \frac{-1}{h_i} \quad (\text{in case of non-uniform mesh}) \\ &= \frac{-1}{h} \quad (\text{in case of uniform mesh for which all } h_i \text{ are equal}) \end{aligned}$$

Linear System Formulation (6/8)

Summary: Matrix A and Right-Hand Vector f

Internal mesh points x_i for $2 \leq i \leq N$

- After assembly, the i -th row of the matrix A is given by

$$A[i, :] = \begin{pmatrix} 0 & \dots & 0 & \underbrace{-1/h_{i-1}}_{i-1} & \underbrace{1/h_{i-1} + 1/h_i}_i & \underbrace{-1/h_i}_{i+1} & 0 & \dots & 0 \end{pmatrix}$$

- After assembly, the i -th component of the vector f is given by

$$f[i] = \frac{h_{i-1} + h_i}{2} f(x_i)$$

Linear System Formulation (7/8)

Summary: Matrix A and Right-Hand Vector $\mathbf{f}$

Treatment of the Boundary Conditions

- **Dirichlet boundary conditions:** $u(x = 0) = 0$:

modify first equation in linear system

see finite difference method

- **Neumann boundary conditions:** $u'(x = 1) = \alpha$

add term $\alpha \phi_i(x = 1)$ to vector $\mathbf{f}$

see lab sessions for details

Linear System Formulation (8/8)

Exercise

- assume again a non-uniform mesh on Ω with 4 nodes $x = 0$, $x = 1/3$, $x = 1/2$ and $x = 1$;
- compute matrix A and vector $\mathbf{f}$;
- assume homogeneous boundary conditions at both end points. Compute the solution vector $\mathbf{f}$;
- verify properties of the matrix A

Assembly Element-by-Element of Vector (1/12)

Remember that

- i -th component of the right-hand side vector $\mathbf{f}$ is given by

$$f_i = \int_{x_{i-1}}^{x_{i+1}} f(x) \phi_i(x) dx = \int_{e_{i-1}} f(x) \phi_i(x) dx + \int_{e_i} f(x) \phi_i(x) dx$$

Assembly Element-by-Element of Vector (2/12)

Element e_i contributes to f_i and f_{i+1} only

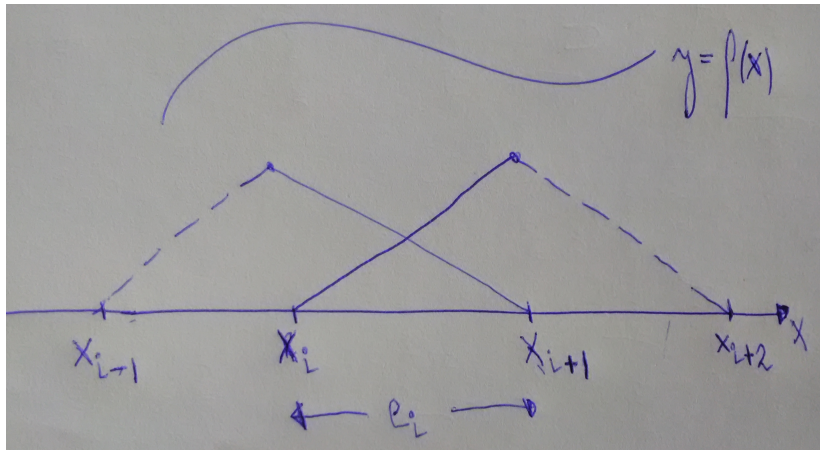
- i -th component of the right-hand side vector $\mathbf{f}$

$$f_i = \int_{e_{i-1}} f(x) \phi_i(x) dx + \int_{e_i} f(x) \phi_i(x) dx$$

- $i + 1$ -th component of the right-hand side vector $\mathbf{f}$

$$f_{i+1} = \int_{e_i} f(x) \phi_{i+1}(x) dx + \int_{e_{i+1}} f(x) \phi_{i+1}(x) dx$$

Assembly Element-by-Element of Vector (3/12)



Assembly Element-by-Element of Vector (4/12)

How does element e_i contribute to the vector $\mathbf{f}$?

- we here fix element e_i (rather than fixing component f_i)
- element $e_i = [x_i, x_{i+1}]$ sees shape functions $\phi_i(x)$ and $\phi_{i+1}(x)$
all other basis functions are zero on the element e_i
- the element e_i therefore contributes to **two** component of $\mathbf{f}$
these components are f_i and f_{i+1}
(same indices as basis functions)
- this leads to an element-by-element construction of vector $\mathbf{f}$

Assembly Element-by-Element of Vector (5/12)

How does element e_i contribute to the vector $\mathbf{f}$?

- $f_{e_i} \in \mathbb{R}^2$ contribution of element e_i to global vector $\mathbf{f}$

$$f_{e_i} = \begin{pmatrix} \int_{e_i} f(x) \phi_i(x) dx \\ \int_{e_i} f(x) \phi_{i+1}(x) dx \end{pmatrix}$$

- use trapezoidal rule of integration (see earlier)

$$f_{e_i} = \begin{pmatrix} \int_{e_i} f(x) \phi_i(x) dx \\ \int_{e_i} f(x) \phi_{i+1}(x) dx \end{pmatrix} = \frac{h_i}{2} \begin{pmatrix} f(x_i) \\ f(x_{i+1}) \end{pmatrix}$$

- given mesh Ω^h and source $f(x)$, f_{e_i} for each e_i can be computed

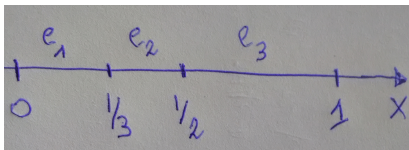
Assembly Element-by-Element of Vector (6/12)

Finite Element Assembly of the Vector $\mathbf{f}$

- loop over all of the N elements e_i in the mesh Ω^h
- on e_i compute the local element vector $\mathbf{f}_{e_i} \in \mathbb{R}^2$
the local element vector has **two** components
- add local element vector to the global vector $\mathbf{f} \in \mathbb{R}^{N+1}$
the global vector $\mathbf{f}$ has **$N + 1$** components
- $\mathbf{f} = \mathbf{f} + \mathbf{f}_{e_i}$
assembly requires taking the mesh connectivity into account
connectivity here between nodes x_i and the elements e_i into account

Assembly Element-by-Element of Vector (7/12)

Assembly for Three-Element Mesh



$$f = \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}$$

A vertical line segment representing a 2D domain. The domain is divided into three elements by nodes at $y = 0$, $y = 1/3$, $y = 1/2$, and $y = 1$. The elements are labeled e_1 , e_2 , and e_3 to the left of the line. The nodes are marked with dots and labeled with their corresponding y values to the right of the line.

Assembly Element-by-Element of Matrix (8/12)

How does element e_i contribute to the vector A ?

- element $e_i = [x_i, x_{i+1}]$ sees shape functions $\phi_i(x)$ and $\phi_{i+1}(x)$
- the element e_i therefore contributes to **four** component of A
these components are A_{ii} , $A_{i,i+1} = A_{i+1,i}$ and $A_{i+1,i+1}$

Assembly Element-by-Element of Matrix (9/12)

How does element e_i contribute to the vector A ?

- $A_{e_i} \in \mathbb{R}^{2 \times 2}$ contribution of element e_i to global vector A

$$A_{e_i} = \begin{pmatrix} \int_{e_i} \phi'_i(x) \phi'_i(x) dx & \int_{e_i} \phi'_i(x) \phi'_{i+1}(x) dx \\ \int_{e_i} \phi'_i(x) \phi'_{i+1}(x) dx & \int_{e_i} \phi'_{i+1}(x) \phi'_{i+1}(x) dx \end{pmatrix}$$

- use derivative of the shape functions

$$A_{e_i} = \frac{1}{h_i} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

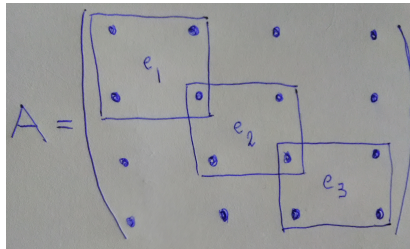
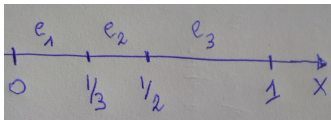
Assembly Element-by-Element of Matrix (10/12)

Finite Element Assembly of the Matrix A

- loop over all of the N elements e_i in the mesh Ω^h
- on e_i compute the local element matrix $A_{e_i} \in \mathbb{R}^{2 \times 2}$
the local element matrix has **two by two** components
- add local element matrix to the global matrix $A \in \mathbb{R}^{(N+1) \times (N+1)}$
the global matrix A has **$N + 1$ by $N + 1$** components
- $A = A + A_{e_i}$
assembly requires taking the mesh connectivity into account
connectivity here between nodes x_i and the elements e_i into account

Assembly Element-by-Element of Matrix (11/12)

Assembly for Three-Element Mesh



Implementation in Julia (12/12)

- see fourth lab session

Exercises

- to be defined